Directions

The following is a template for you to write your research paper with. For every section there are questions that you should address in your answer and a word count. **Please treat these word counts as a minimum.** But you may go over the word count if you please – but do not go under it. To check a word count in Google Docs, highlight the portion and press Ctrl + Shift + C, or ⌘ + Shift + C for Mac users.

Certain sections are marked with a star symbol (★), which means you are required to cite multiple sources in your response. Sections with no ★ do not require an external source but you are welcome to cite one if you think it is appropriate.

Note that you are writing a proper research paper here, so keep your word choice professional, proper, and formal. Don't use contractions in your writing. Using "we" is appropriate for this type of paper, but don't use "I" or "you."

When citing texts, you *must* follow academic plagiarism rules. Do *not* directly copy-paste from another source without quoting or paraphrasing. Failure to do so will cause significant trouble for you. See the next page for a more thorough explanation of how to properly do it.

For reference, you may look at previous research papers to get a feel on how to write it (please note that these papers do not have the exact same format as the one that you are writing; it is roughly similar).

- Paper 1 - Volunteer Management
- Paper 2 - COVID Tracking
- Paper 3 - Blood Sugar Tracking
- Paper 4 - Marine Pollution (Same format as this template!)

If you're not terribly fond of writing or it's not a strong point for you, then your instructor should be able to provide writing assistance and guide you on how to write your ideas down. Do note however that you are expected to write a significant portion of this paper by yourself.

Citations and Sourcing

Citations in academic research papers are different from how you would normally do it in a typical research paper. Unlike your typical paper where you must include the author(s), date published, and specific page, all that is required for a citation is the index of the work cited at the end of the sentence in square brackets. That means it's easier to cite, but you still must follow proper citation guidelines. Try not to copy-paste sections from other works. If you do, put it in quotation marks. And for the record – yes, copy-pasting even 3 words without quotation counts as plagiarism.

Finding Works

Generally stick to Google Scholar for your sources. Don't use works outside of GS, unless you're using a government website (for example, the CDC). When it comes to your works cited, cite in APA format. If you're using Google Scholar to locate your sources, almost all papers found on GS should have an APA style citation for you.

Paraphrasing

Paraphrasing is summarizing or rewording an author's point. It doesn't require quotation marks but still requires citation since you are directly referencing their ideas. Consider changing the whole sentence structure, re-contextualizing ideas, or substituting words with synonyms.

It could be argued that video game players compete so aggressively for mostly self-satisfactory purposes. In one study, players reportedly gained more motivation and satisfaction when they found that they were doing well and did not require as much assistance [2]. In that sense anti-social behavior in video games may not be necessarily linked to violent content.

Quotes

For academic papers, you should actually not quote very much, but you still can if you please. When quoting, only quote a portion that best represents a singular idea or point. Of course, remember to enclose it in quotation marks. If your quote is at the end of the sentence, put your bracket citation after the ending quote, but before the period.

It could be argued that video game players compete so aggressively for mostly self-satisfactory purposes. At least one study has shown that "enjoyment, value, and desire for future play were robustly associated with the experience of autonomy and competence in gameplay" [2]. In that sense anti-social behavior in video games may not be necessarily linked to violent content.

Preliminaries

Title

A [SMART ADJECTIVE] system/program to [PURPOSE] using [TECHNOLOGIES]

For example: *An Intelligent Mobile Application to Assist in Video Editing and Human Motion Tracking using Machine Learning and Object Detection*

Tivvy : Voice Activated Productivity Timer

Keywords

Timer
Voice Activation
Voice Recognition
Machine Learning

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Section 1 – Introduction (500 words)

1.1 – Introduction to Problem (250 words) ★

What problem are you trying to solve? What's the background/history of your problem? Why is your problem important? Who does your problem affect in the long run? Provide some statistical examples.

- Productivity
- Easy Voice Interaction
- Offline Solution
    - Privacy concerns from other "smart" devices

1.2 – Method Proposal (250 words)

Propose your method of solving this problem. First, explain what your solution is in one sentence. Then, discuss how your solution solves your problem. Why do you think it is an effective solution, and why would it be a better solution than other methods that you've discussed?

- Offline Timer Device
- Edge devices (Nicla)
- Intuitive UI

Tivvy is a standalone voice activated offline timer that is able to take in voice commands to create a variety of task related timers along with visual animations and sound effects to help users keep track of their work throughout the day. We set out to create a device that prioritizes offline functionality both in the name of privacy and general convenience. This is done through the use of edge technologies like the Nicla Voice chip that pairs a microphone with a custom trained AI model to be able to identify commands in a variety of conditions, which is then paired to a Qualia board that powers the display and speakers to interact with the user. Users can set up to three different timers running at any given time and view rings to see how much time they still have remaining. The entire system is packaged into a compact enclosure with the option of either directly plugging in the device to the wall with one cord or through a battery instead.

Section 2 – Challenges (300 words)

2.1 - Challenge A (100 words)

Think of a major component of your program. Imagine the sorts of problems that you had to consider when implementing this component. What could potentially cause problems that you need to address? Provide a short explanation as to how you could resolve them. Do not go into detail about how you solved it in your project! Word it as if you would be discussing how you would go about solving these objections (instead of “I used,” say “I could use”).

- Voice Recognition
  - Offline & edge requirements
  - Keyword spotting (data acquisition)
  - Solution: Nicla Voice Chip + Model Training

2.2 – Challenge B (100 words)

Follow the same directions as 2.1 but with a different component in mind.

- Enclosure design & Parts Organization
  - Bespoke case design
  - Being able to fit a display, battery, and the two chips in as small of a footprint as possible
  - 3D CAD Program (FreeCAD)

2.3 – Challenge C (100 words)

Follow the same directions as 2.1 but with a different component in mind.

- Sound design
  - Speaker Audio Processing limitations
  - Mp3 translated into raw data
  - When to play
  - Gather specific sounds needed for the intended event + an mp3 to data translation script to automate audio additions

If you’re confused, refer to the next page for a quick guide on how to write this out.

Section 2 Writing Guide

Consider a mobile application with an AI that can detect specific objects. 3 components would be:

- UX (user interface and experience)
- AI Model
- Data Storage and Tracking

Let's consider problems that would come up with the second component:

- Where are you going to get data from? —> collecting data takes a lot of time
- How robust does this AI need to be? —> the AI can't be too big or too small
- What if the AI is not accurate? —> the AI can't be inconsistent

How would we solve these problems?

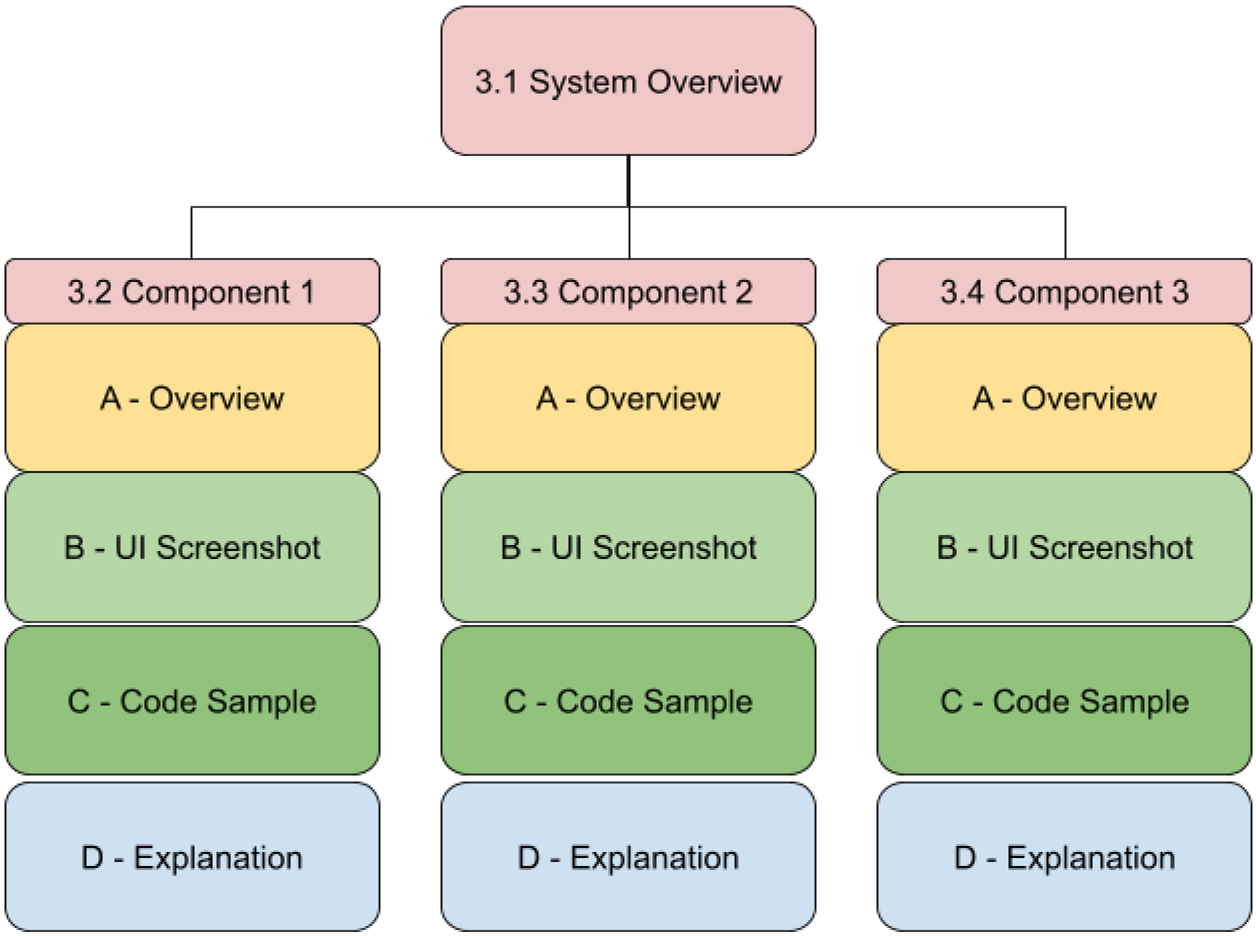
- We will get data using a data scraper.
- Our AI will at minimum focus on identifying basic foods like bread and rice.
- To ensure the AI is accurate, we will train it with at least 500 images and perform quantitative assessments.

At this point, you will only have to write it out. The following exceeds the 100 word minimum (132).

The performance of this application is heavily impacted by the artificial intelligence model that we are using. We must first consider where we are going to find adequate data, as AI models need hundreds of images to train. To expedite this process, we will use an image scraper on Flickr. The AI must also be comprehensive, but it cannot be so large that development would take too long, but not so small that it serves no utility. We will establish that at minimum our AI must be able to identify basic foods such as bread and rice. Our AI must also be accurate at identifying these objects to provide utility to users and so they are not frustrated. To account for this, we will train at least 500 images per object and perform quantitative assessments for adjustment.

Section 3 – Method Analysis (850 words)

Your Section 3 should be based on this structure below:



3.1A – Diagram

Provide a flowchart of your program. You should probably already have one made for you. If not, construct a diagram with your application screens. If the one made for you isn't indicative of the final product, modify it to be so.

3.1B – System Overview (250)

Explain the main structure of your program. What are 3 major components that your program links together? Discuss the “flow” of your program – how does your program work from start to finish? What did you use to make this program?

1. Voice Recognition System
2. Hardware (Screen/Qualia, etc)
3. Software

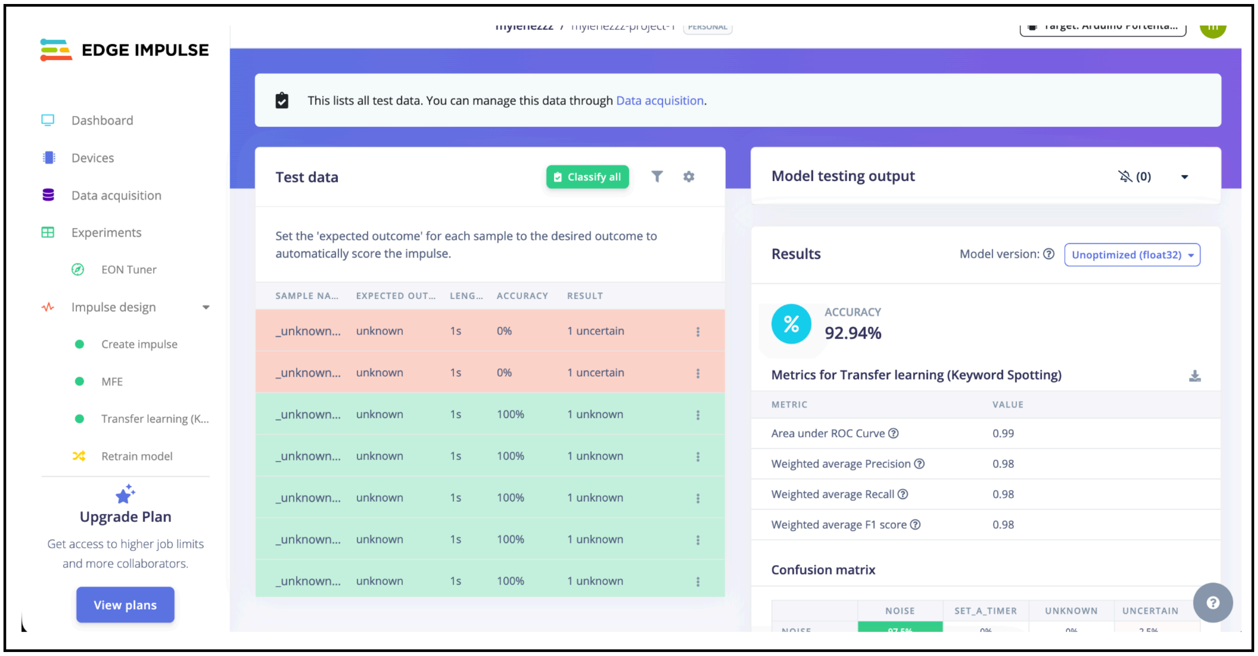
3.2A – Component Analysis A (50 words)

Take one of the components that you established in 3.1. Discuss what the component’s purpose is. What services did you use to implement this system? Does your component rely on a special concept (eg. NLP, Neural Networks, Authentication)? If so, briefly explain what the concept is. In a very broad sense how does the component function with regards to your program?

Voice Recognition System
- Nicla voice chip
- Edge Impulse dataset & training
- Command recognition

3.2B – UI Screenshot

Show a screenshot of your program that best displays your program. If necessary, you can add an additional screenshot if you feel it best communicates what your component does.



3.2C – Code Sample

Find a screenshot of your code that best represents the component. It does not need to necessarily include the entire component's code. It's best if you can find a method that best encapsulates your component, but fine if not.

```
#include <Arduino.h>
#include <NDP.h>
#include <ArduinoBLE.h>
#include <Nicla_System.h>   // for nicla::leds + color constants

// ======================= CONFIG =======================
#define DEBUG_SERIAL 1
#define LED_FEEDBACK 1

#define BLE_SCAN_RESTART_MS 2500
#define BLE_CONNECT_TIMEOUT_MS 20000
#define COMMAND_TIMEOUT_MS 1500
#define MAX_TOKENS 10

// ======================= NUS UUIDs (MUST MATCH QUALIA) =======================
#define SERVICE_UUID             "6E400001-B5A3-F393-E0A9-E50E24DCCA9E"
#define CHARACTERISTIC_UUID_RX   "6E400002-B5A3-F393-E0A9-E50E24DCCA9E" // Qualia RX (we WRITE here)
#define CHARACTERISTIC_UUID_TX   "6E400003-B5A3-F393-E0A9-E50E24DCCA9E" // Qualia TX (optional notify)

static const char* TARGET_DEVICE_NAME = "TimerDevice"; // Qualia advertises NimBLEDevice::init("TimerDevice")

// ======================= TOKEN IDS (must match your model/templates) =======================
enum TokenId : uint8_t {
 TOK_SET = 0,
 TOK_CANCEL = 1,
 TOK_ADD = 2,
 TOK_MINUS = 3,
 TOK_STOP = 4,
 TOK_TIMER = 5,
 TOK_MINUTE = 6,
```

```
TOK_MINUTES = 7,
TOK_HOUR = 8,
TOK_HOURS = 9,
// Numbers 1-19
TOK_ONE = 10,
TOK_TWO = 11,
TOK_THREE = 12,
TOK_FOUR = 13,
TOK_FIVE = 14,
TOK_SIX = 15,
TOK_SEVEN = 16,
TOK_EIGHT = 17,
TOK_NINE = 18,
TOK_TEN = 19,
TOK_ELEVEN = 20,
TOK_TWELVE = 21,
TOK_THIRTEEN = 22,
TOK_FOURTEEN = 23,
TOK_FIFTEEN = 24,
TOK_SIXTEEN = 25,
TOK_SEVENTEEN = 26,
TOK_EIGHTEEN = 27,
TOK_NINETEEN = 28,
// Tens
TOK_TWENTY = 29,
TOK_THIRTY = 30,
TOK_FORTY = 31,
TOK_FIFTY = 32,
TOK_SIXTY = 33,
TOK_SEVENTY = 34,
TOK_EIGHTY = 35,
TOK_NINETY = 36,
// Timer names
TOK_BAKING = 37,
TOK_COOKING = 38,
TOK_BREAK = 39,
TOK_HOMEWORK = 40,
TOK_EXERCISE = 41,
TOK_WORKOUT = 42,
TOK_UNKNOWN = 255
};
```

```
// Token name lookup (what your NDP model should output as labels)
static const char* TOKEN_NAMES[] = {
 "set", "cancel", "add", "minus", "stop", "timer",
 "minute", "minutes", "hour", "hours",
 "one", "two", "three", "four", "five", "six", "seven", "eight", "nine",
 "ten", "eleven", "twelve", "thirteen", "fourteen", "fifteen",
 "sixteen", "seventeen", "eighteen", "nineteen",
 "twenty", "thirty", "forty", "fifty", "sixty", "seventy", "eighty", "ninety",
 "baking", "cooking", "break", "homework", "exercise", "workout"
};
static const int TOKEN_COUNT = (int)(sizeof(TOKEN_NAMES) / sizeof(TOKEN_NAMES[0]));

// Timer name strings (must match what Qualia expects)
static const char* TIMER_NAMES[] = { "Baking", "Cooking", "Break", "Homework",
"Exercise", "Workout" };
static const int TIMER_NAME_COUNT = 6;
```

3.2D – Code Explanation (150 words)

To put it simply, what's going on in that screenshot? When does this code run in your program? What does each method represent? What variables are being made? Walk through each method that you showed step by step. If it's communicating with a backend server, what's the server doing?

3.3A – 3.3C – Component B (200 words)

Repeat the same steps as Steps 3.2A – 3.2C on a different component.

Hardware:
- Qualia S3 RGB666
    - Powers the display & command execution & general timer logic
    - Speaker audio playback control
- Nicla Voice Chip
    - Keyword spotting & command recognition
- Bluetooth interplay
- Case with power split to run both chips
    - Ability to run wired/wireless

```
// ====================== RENDER TIMERS ======================
static void renderTimers() {
 if (active_timer_count == 0) {
   drawNoTimersScreen();
   return;
 }

 // Get active timer indices
 int activeIndices[MAX_TIMERS];
 int count = 0;
 for (int i = 0; i < MAX_TIMERS && count < active_timer_count; i++) {
   if (timers[i].active) {
     activeIndices[count++] = i;
   }
 }
 gfx->fillScreen(COLOR_IDLE_BG);
 uint32_t now = millis();

 for (int i = 0; i < count; i++) {
   int idx = activeIndices[i];
   PanelLayout layout = panel_layout_for(active_timer_count, i);
   draw_timer_panel(layout, timers[idx], now);

   char hhmmss[9];
   fmt_hhmmss(timers[idx].seconds_left, hhmmss);
   strcpy(last_text[i], hhmmss);
   strcpy(last_name[i], timers[idx].name);
 }
}
```

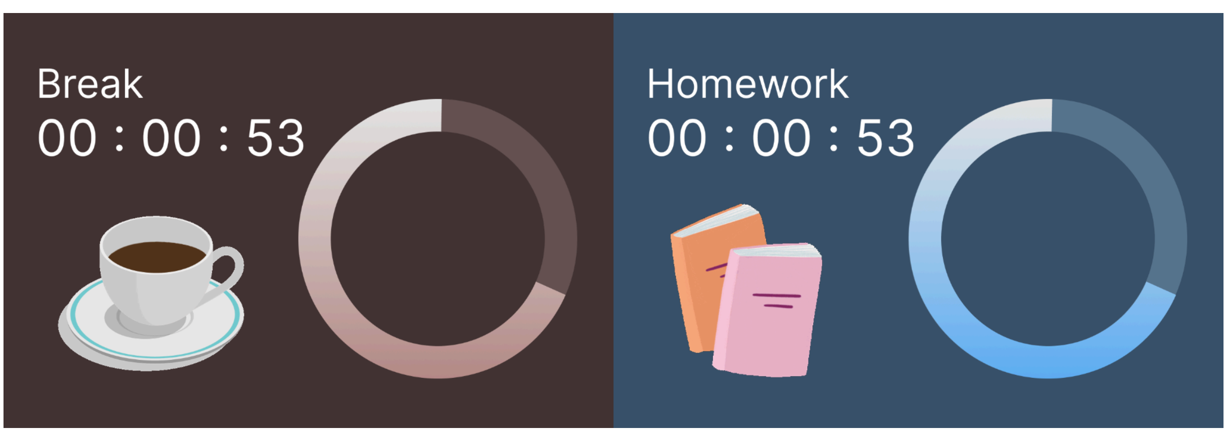
```
static void draw_timer_ring(const PanelLayout& layout, const Timer& timer, float
frac, uint32_t now) {
 #if USE_RING
 TimerTheme theme = resolved_theme_for_timer(timer);
 bool flash = ((now / 250) % 2) == 0;
 uint16_t ring_start = timer.ringing
   ? (flash ? COLOR_ALERT_START : lerp565(COLOR_ALERT_START, COLOR_ALERT_ORANGE,
80))
   : theme.ring_start;
 uint16_t ring_end = timer.ringing
   ? (flash ? COLOR_ALERT_RED : COLOR_ALERT_ORANGE)
```

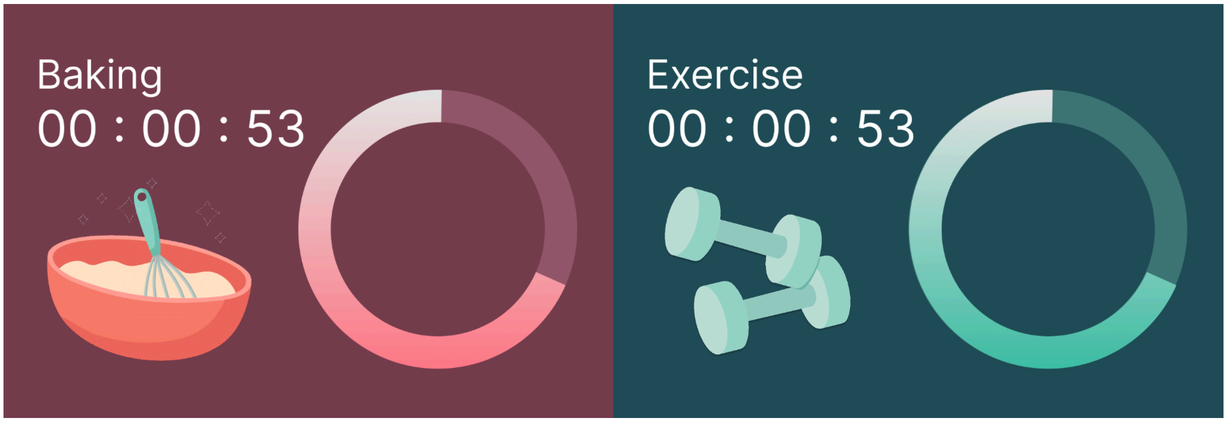
```
  : theme.ring_end;
uint16_t ring_empty = timer.ringing
  ? lerp565(theme.ring_empty, COLOR_ALERT_ORANGE, 80)
  : theme.ring_empty;
float draw_frac = timer.ringing ? (flash ? 0.04f : 0.12f) : frac;
int ring_size = ring_size_px(layout.ring_scale);

ensure_ring_lut(layout.ring_scale);
draw_ring(draw_frac, CAP_LEAD, ring_start, ring_end, ring_empty,
          layout.ring_cx - ring_size / 2,
          layout.ring_cy - ring_size / 2,
          theme.bg, layout.ring_scale);
#endif
}
```

3.4A – 3.4C – Component C (200 words)

Repeat the same steps as Steps 3.2A – 3.2C on a different component.

Software
- 1-3 timers that can be set for different activities
- UI embellishments like the animated gifs & ring
- Command recognition & processing
    - Briefly mention the available commands
    - Set, Stop, Add, Minus, Cancel
- Bluetooth interop






```
static ParsedCommand parseCommand(const std::string& msg) {
 ParsedCommand result = {CMD_NONE, "", 0};
  if (msg.find("CMD:SET") != std::string::npos) {
   result.cmd = CMD_SET;
 } else if (msg.find("CMD:CANCEL") != std::string::npos) {
   result.cmd = CMD_CANCEL;
 } else if (msg.find("CMD:ADD") != std::string::npos) {
   result.cmd = CMD_ADD;
 } else if (msg.find("CMD:MINUS") != std::string::npos) {
   result.cmd = CMD_MINUS;
 } else if (msg.find("CMD:STOP") != std::string::npos) {
   result.cmd = CMD_STOP;
   return result;
 }
  // Parse NAME
 size_t namePos = msg.find("NAME:");
 if (namePos != std::string::npos) {
   size_t start = namePos + 5;
   size_t end = msg.find(",", start);
   if (end == std::string::npos) end = msg.length();
   std::string name = msg.substr(start, end - start);
   strncpy(result.name, name.c_str(), 15);
   result.name[15] = '\0';
 }
  // Parse DURATION
 size_t durPos = msg.find("DURATION:");
 if (durPos != std::string::npos) {
   size_t start = durPos + 9;
   result.duration = atoi(msg.substr(start).c_str());
 }
```

```
  return result;
}
```

If you're confused, refer to the next page for a quick guide on how to write this out.

Section 3 Writing Guide

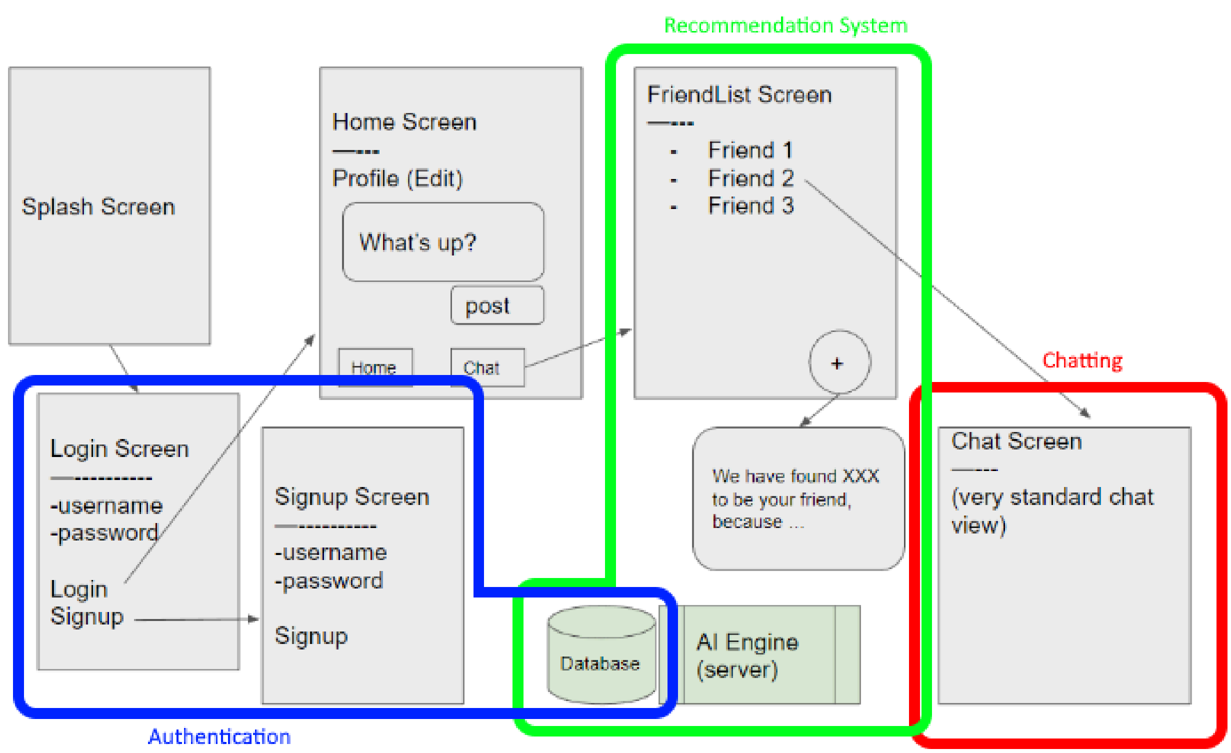
Suppose we are writing for a social networking application that will let users find and send messages to one another. Let’s say that we’ve established that our main components are the chatting system, the authentication backend, and a recommendation system that uses AI.

3.1 - System Overview

Main Structure:

- Three parts: an authentication service, a recommendation system, and a chatting system
- Information is all uploaded and stored to Firebase database

Here’s our flow chart. Let’s circle the 3 major components that we can identify. For 3.1A, you would put the flow chart without these circles.



Flow is as simple as following the arrows.

- Get splash screen
- Login Screen
- Signup Screen if needed
- Both go to the dashboard
  - you can make posts from the dashboard
  - displays profile information
- Friend Screen
  - shows all your friends that you can talk to
  - if you don’t have any, then ask for a recommendation

    - AI server will respond back with a potential new friend
- Chat Screen
    - chat with someone
    - similar to SMS or WhatsApp

What did you use to make this program?

- Flutter
- Firebase
- Repl.it

The following exceeds the 250 minimum word limit (273 words)

The program is split into three large components: an authentication service, a recommendation system, and a chatting system. All the information sent through the app is stored within a database to later be processed by a sentiment analysis AI. To construct this program, we used Flutter for the codebase, Firebase for authentication and database services, and Repl.it to host our sentiment analysis AI model. We decided to use Firebase primarily because of its ease of use but also because we can allow certain portions of our database to be accessible through HTTP requests. We decided to host our sentiment analysis AI on a repl.it server running flask because it is simple to set up and make API requests to. Our program is designed to be a simple application that users can download free on app stores. When opened, it will display a splash screen to the user and prompt them to log in. If the user doesn't have an account, they may sign up. Logging in and signing up both use the authentication service offered by Google's Firebase, which allows us to manage a database efficiently. When they log in they are shown the dashboard, and can make posts that are uploaded to Firebase, or they can visit a Friends List screen. If they do not have friends in the system, they can prompt the app for a friend. The app will send an HTTP request to a server hosting a sentimentality AI that will determine the best match for a user. There is also a chat screen for users to communicate through that operates similarly to SMS or WhatsApp messaging.

3.2A – Recommendation System

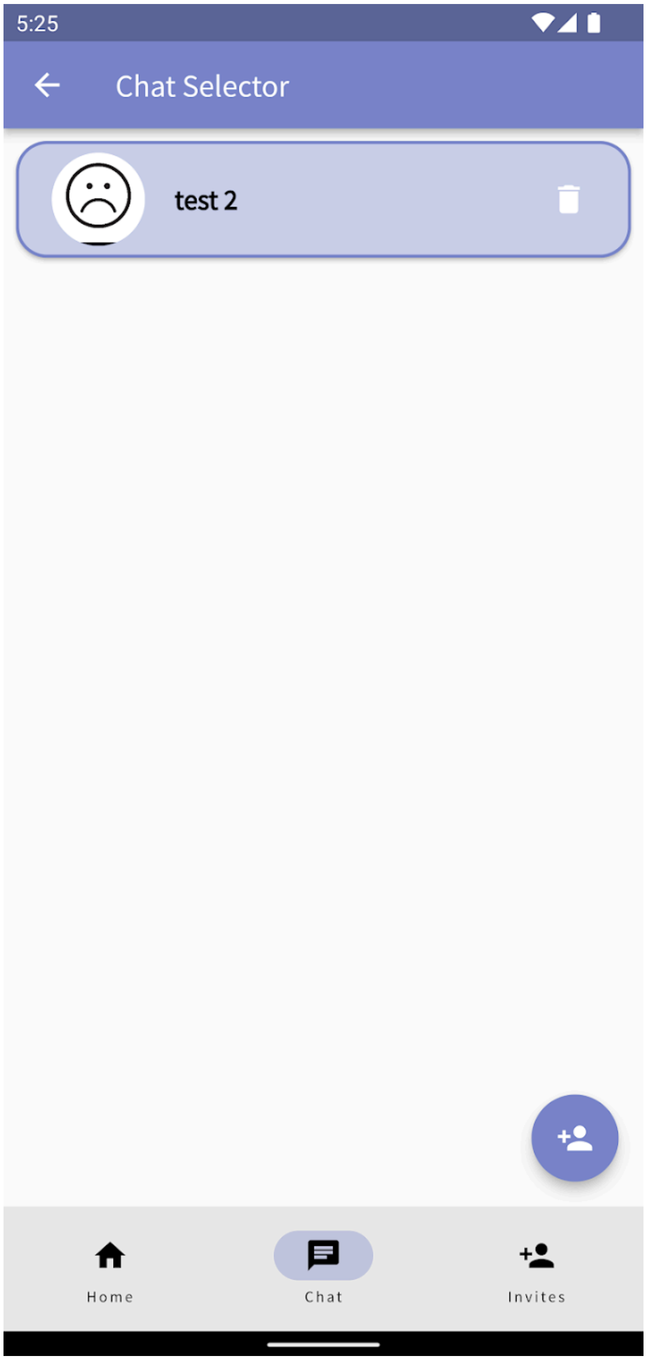
Let’s try writing about the third component.

- What is its purpose?
    - the “social” component of the app, allows users to connect with one another
- What services did you use?
    - Repl.it server
- Explain special concepts
    - Natural Language Processing
        - a way of computing language as data
        - ability for machines to recognize human emotions and suggestions through written words only
- Connect to program
    - To be used in conjunction with friend request system to find new users to chat with, and is required in order to instigate a conversation

The following exceeds the 50 word count minimum (76 words)

A key part to the app’s functionality is our recommendation system. This system is integral to the app’s core design because it is the primary way for users to find new users to send friend requests to. This recommendation system relies on a simple repl.it backend server, that hosts an NLP model. NLP, or Natural Language Processing, is a way to compute written language as data and recognize human emotions and suggestions from that data.

3.2B – UI Screenshot



3.2C – Screenshot

We need to find a screenshot that best represents the component and best illustrates how the component is used within the program's codebase. The method that we decided to show here, *sentimentFriendSelector,* includes the code where we make a request to a server, and shows how we process the information that we get back.

```dart
Future<dynamic> sentimentFriendSelector() async {
  String url = "https://userpostsentimentanalysis.marisabelchang.repl.co/" + getUID();
  String pickedUID;
  String pickedName = "";
  String pickedDescription = "";

  //Get a UID from the server
  var response = await http.get(Uri.parse(url));
  print(response.body.toString());
  var listData = jsonDecode(response.body.toString());

  pickedUID = listData[0].toString();
  if (pickedUID == getUID())
    {
      pickedUID = listData[1].toString();
    }
  print("RESULT: " + listData.toString());

  //Get the username and description
  await FirebaseDatabase.instance.ref().child("userProfile").child(pickedUID).once()
      .then((event) {
    print("Successfully grabbed profile for user " + pickedUID);
    var profile = event.snapshot.value as Map;

    pickedName = profile["Username"];
    pickedDescription = profile["Description"];

    print(pickedName);
    print(pickedDescription);
  }).catchError((onError) {
    print("Could not grab user profile for user " + pickedUID);
  });

  return [pickedUID, pickedName, pickedDescription];
}
```

3.2D – Code Explanation

- When does it run in the program?
  - On a friend recommendation page, when the user presses a recommend friend button
- What does the method represent?
  - sentimentFriendsSelector will make an HTTP request to the server hosting the NLP AI, and then display to the user the profile information of the user that the server sent back
- What variables are being made?
  - url – the URL of the HTTP request. To request the server for a recommendation for a specific user, we attach their unique identification at the end of the URL.
  - pickedUID – the user identification of the user that the server sends back. Used to lookup the user's profile information
- What does the server do
  - AI model will go through all user's posts and perform sentiment analysis on each post that they've made
  - It computes the average sentiment per post for that user
  - It does the same for every other user
  - It returns the UID of the user whose average is closest to our user
- Step by step process
  - 1. Generate a URL with the user's UID.
  - 2. Make an HTTP request with that URL. Wait until response
  - 3. With UID, look up Firebase Database and get corresponding user profile info
  - 4. Return an array with the UID, name, and description
  - 5. Data is shown as a pop up and user can either add them as a friend or ask for new recommendation

The following exceeds the 150 word count minimum (287 words)

The method *sentimentFriendSelector* best represents this component. It is called in the program on a recommended friend page when the user presses a button. When called, it will make a request to a separate server hosting an NLP model, and it will then display to the user the profile information of a recommended friend. There are two important variables. "url" is the URL of the HTTP request. To request the server for a recommendation for a specific user, we attach their unique identification at the end of the URL. “pickedUID” is the user identification of the user that the server sends back. It is used to lookup the user’s profile information. When the method runs, it first makes an HTTP request with the URL. The method waits until the data is sent back. Serverside, the AI model will go through all user’s posts and perform sentiment analysis on each post that they’ve made. It computes the average sentiment per post for that user, with lower scores representing more negative posts and higher scores representing more happy posts. It does the same for every other user and returns the UID of the user whose average is closest to our user. Once the server sends a UID back, we search our Firebase Database in the userProfile section and use that UID to access the profile information of the recommended user. Once all the data is compiled, we wrap it up in an array and return it. In the program, this data is then shown to the user via a pop up and the user is given a choice to either request a new recommendation, or accept it, and send a friend request.

Section 4 – Experiments (560 words, 280 per experiment)

4.1A – Experiment A (30 words)

Provide a possible blind spot in your program that you want to test out (for example, an AI's accuracy). Why is it important that this part of your program works well?

Perhaps the most important blindspot is going to be the command accuracy based on different environmental factors since it is the main method of interaction with the device.

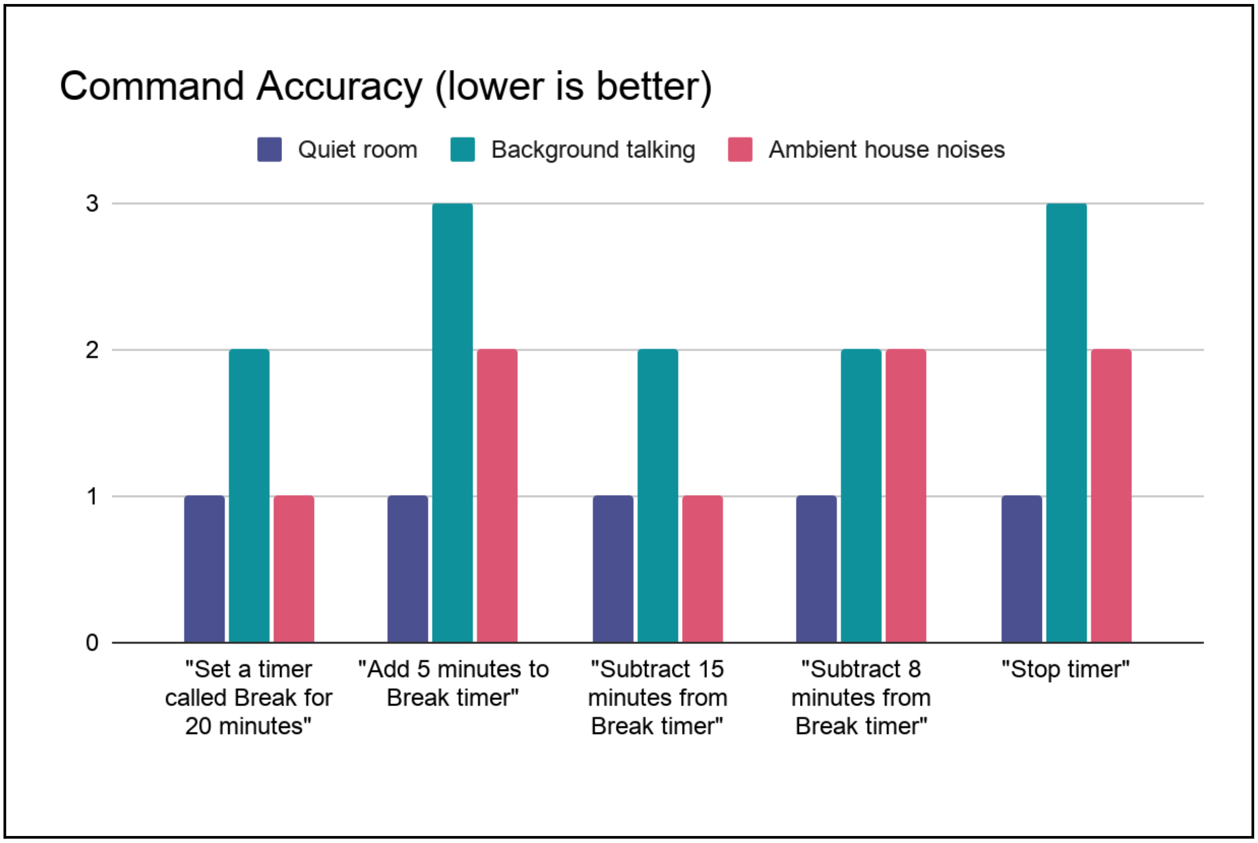
4.1B – Design (100 words)

Explain how you will set up the experiment to test it. Why is the experiment set up that way? State where you're sourcing control data from, if at all.

- Control: quiet room
- Loud environment
  - Background Talking: prerecorded for the sake of reproducibility
  - Ambient house noises (air conditioning, lawn mower, washing machine, vacuum) also prerecorded
- Procedure:
  - Have the device powered on and the person saying commands be a meter away
  - To simulate each of the environments, there will be a speaker emitting the necessary audio 10ft (3 meters) away
  - Command Sequence each time (wait 3 seconds between command, if a command fails repeat two more time before calling end of run):
    - "Set a timer called Break for 20 minutes"
    - "Add 5 minutes to Break timer"
    - "Subtract 15 minutes from break timer"
    - "Subtract 8 minutes from break timer"
    - Wait for ringtone to trigger
    - "Stop timer"
  - Go through the command with each environment type

4.1C – Data and Visualization

Provide the data from your experiment, as a graph.



4.1D – Analysis (150 words)

Analyze the data. What's the mean and median? What's the lowest value? What's the highest value? What data surprised you or did not meet your expectations? More importantly, why do you think that it turned out that way? What has the biggest effect on your results?

4.2A – 4.2D – Experiment 2 (280 words)

Repeat the steps for 4.1 on another potential blind spot.

Battery Life based on environment conditions and usage patterns
Environment: room temp, hot/colder environments - Hot environment: in the sun - Cold environment: fridge Procedure: - Get the current battery charge at the beginning of the run

- Set a timer on the device for 5 minutes
- Check the remaining battery once the timer ends

Device State:

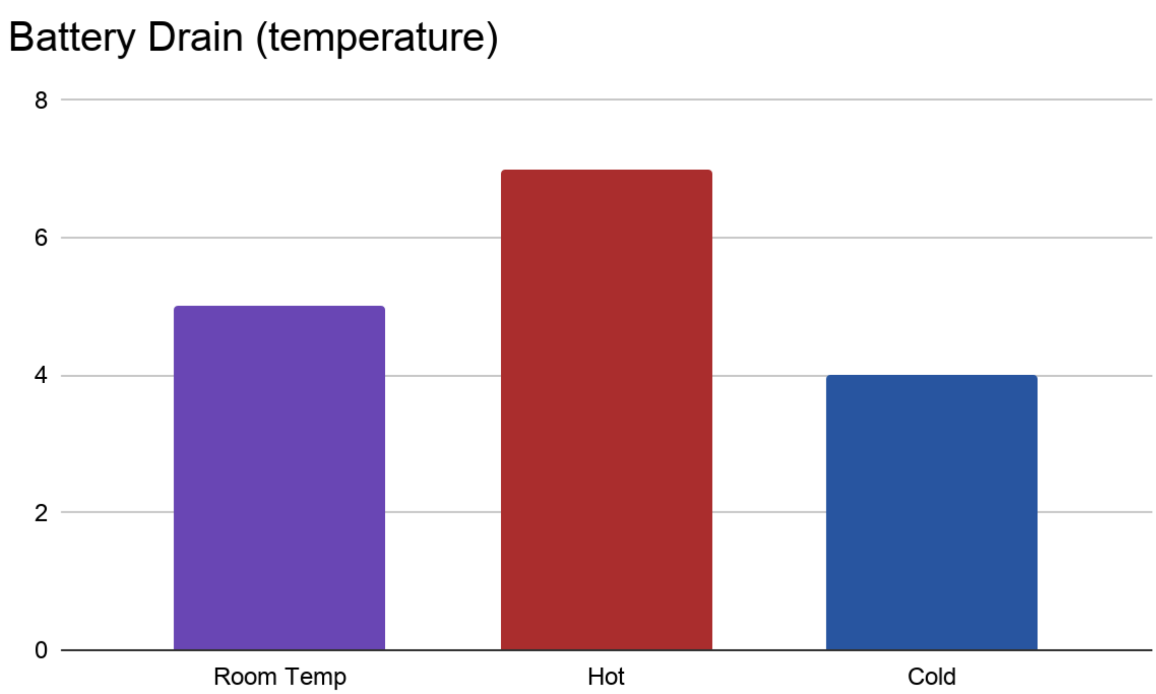
Test these conditions:

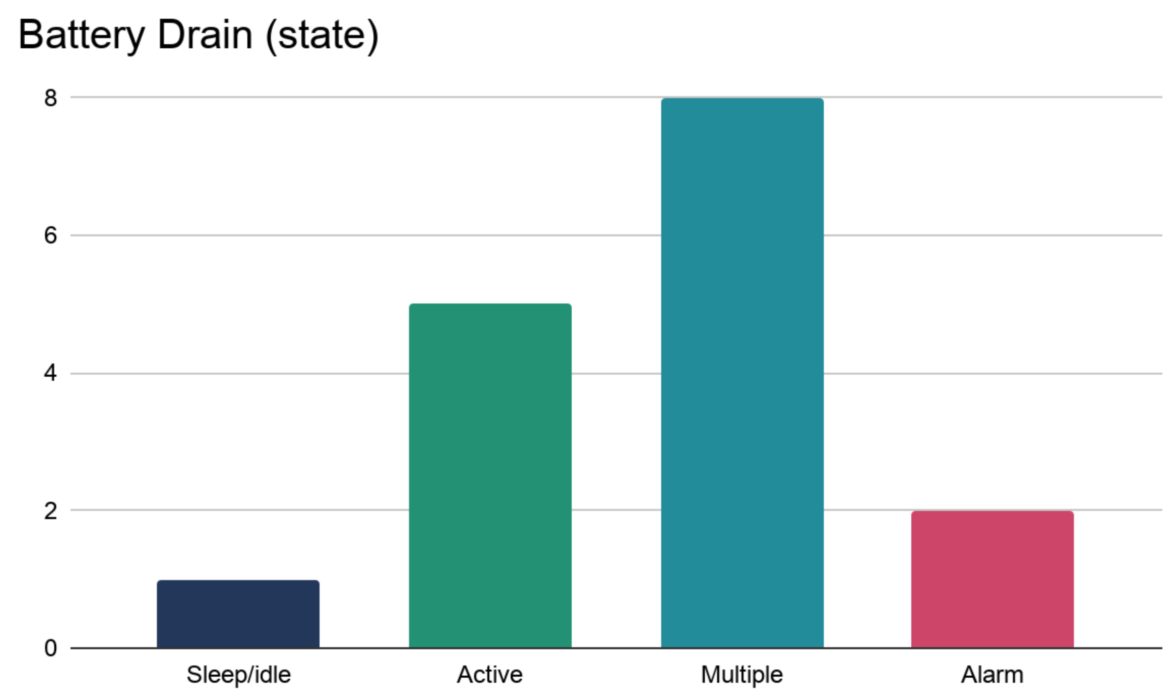
1. Sleep/idle state: no timer running, device allowed to auto-sleep.
2. Active timer state: one 5-minute timer running with the display on.
3. Multiple timer state: three timers running at once.
4. Alarm state: timer finished, ringtone/sound/visual alert active.

Procedure:

Start each test at the same battery percentage, maybe 100% or 80%. Run the device in one condition for a fixed time, like 10 or 15 minutes, then record the ending battery percentage. Repeat each condition 2-3 times and graph average battery loss per condition.

- How often commands are issued
- Number of timers set
- Note: device auto-sleeps if no timer for 3 minutes





Section 5 – Methodology Comparison (300 words)

5.1 – Methodology A (100 words) ★

Find a scholarly source that tries to tackle the same problem that you're solving. Provide an explanation on how the solution works. How effective is this solution? What are its limitations? What are some things that it ignores? What does your project do that improves on what they tried?

Framework For Preventing Procrastination And Increasing Productivity | IEEE Conference Publication | IEEE Xplore. (n.d.). Ieeexplore.ieee.org. https://ieeexplore.ieee.org/abstract/document/9451773/

- ProScore: point score and leaderboard app
- Flutter & Firebase
- Limitations: requirement to have a smartphone; also potential for distraction by the phone itself. Honor system, more thorough monitoring would be needed otherwise
- Hardware dedicated to a specific and focussed productivity subset

5.2 – Methodology B (100 words) ★

Repeat the same steps as 5.1.

McDougall, D., Morrison, C., & Awana, B. (2012). Students with disabilities use tactile cued self-monitoring to improve academic productivity during... ResearchGate, 39, 119–130. https://www.researchgate.net/publication/308029011_Students_with_disabilities_use_tactile_cued_self-monitoring_to_improve_academic_productivity_during_independent_tasks

- MotivAider: vibration device in user's pocket
- Periodically vibrate to remind users to check if they still have work to do
- Effective during testing for subjects
- Ignores people who don't have a disability
- Limitation: requires the user to actually want to be productive
- Wider audience focus with a focus on accessibility

5.2 – Methodology C (100 words) ★

Repeat the same steps as 5.1.

Uddin, M., Allen, R., Huynh, N., Vidal, J. M., Taaffe, K. M., Fredendall, L. D., & Greenstein, J. S. (2017). Effectiveness of a Countdown Timer in Reducing or Turnover Time. Journal of Mobile Technology in Medicine, 6(3), 25–33. https://www.journalmtm.com/2017/effectiveness-of-a-countdown-timer-in-reducing-or-turnover-time/

- ORTimer: Android app acting a countdown timer in operating rooms to reduce turnover time.
- Effectiveness: shown to work in the Greenville Memorial Hospital where it was tested
- Limitation: wasn't tried yet in other hospitals to determine full strength of reproducibility
- Difference: cheap hardware instead of a mobile app & more general purpose audience as opposed to OR TOT focus

Section 6 – Conclusions (200 words)

6.1 – Limitations and Improvements (150 words)

What are some limitations to your project? What do you think needs to be fixed? How would you implement these if you had more time with your project?

- More refined case
    - Button controls
- More refined voice recognition
- Higher quality screen (adjustment of existing UI palette which was originally made to compensate current screen color accuracy)

6.2 – Concluding Remarks (50 words)

Provide some concluding remarks.

Section 7 – Summaries (200 words)

Papers tend to include subsections with recaps or previews of other sections in the paper. For your convenience, we've compiled these subsections at the end so that you can more easily recap your findings. This section may be tedious, but you will still have to fill these out. Don't copy-paste anything from your previous sections here.

7.1 – Experiment Recap (150 words)

Essentially, recap Section 4 in 150 words. What were you trying to test with each experiment? How did you set up each experiment? What were the most significant findings for each experiment? Briefly recap the reasonings for why your results came out the way they did.

7.2 – Methodology Comparison (150 words, 50 per method)

Recap the 3 methodologies that you went over in Section 5. What does each try to accomplish? What are the shortcomings of the 3 solutions? What did your project do to try and improve on these works?

7.3 - Abstract (150 words)

An abstract for paper writing is a summary of the entire paper, including the introduction, methodology, experiments, and results. In your case, you're basically summarizing the entire paper from Sections 1 to 6, but in 150 words.

Discuss briefly the background to your problem that you're trying to solve. Then, provide a proposal on how you intend to solve this problem. Summarize the key technologies and components of your program. What challenges were there and how did you fix them? Discuss how you applied your application to various scenarios during experimentation. Present the most important results that you found. Why is your idea ultimately something that people should use?

Section 8 - References (15-20 References)

If you've done a considerable amount of research prior to or while writing this paper, then you should already have the links for your sources. Cite in APA for this paper. If you already have the citations ready at this point, great! If not, assuming you consulted Google Scholar for your sources, each source page should have some area where you can copy an APA citation of it and paste it here. If you didn't consult Google Scholar (i.e. you have an article from a government resource or you're using a news article), then you can consult this website to help you generate them quickly. If you don't have your sources down at this point or haven't conducted research, then consult this video as an example on how to quickly allocate your sources. For each reference, place its number in brackets (for example, [2]). Note that the number that you place for each source doesn't reflect when you cite it in the paper. Please remember to go back to where you placed your inline citations to reflect its corresponding source number.

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